

Periodic Properties

Questions with Answer Keys

JEE Main 2020 Chapterwise

Chemistry

Q1 JEE Main 2020 - 2 September (Morning)

In general the property (magnitudes only) that show an opposite trend in comparison to other properties across a period is

- (A) Electron gain enthalpy
- (B) Electronegativity
- (C) Ionization enthalpy
- (D) Atomic radius

Q2 JEE Main 2020 - 2 September (Evening)

Three elements X, Y and Z are in the 3rd period of the periodic table. The oxides of X, Y and Z, respectively, are basic, amphoteric and acidic. The correct order of the atomic numbers of X, Y and Z is

- (A) $x < z < Y$
- (B) $Y < x < z$
- (C) $z < Y < x$
- (D) $x < Y < z$

Q3 JEE Main 2020 - 3 September (Morning)

The atomic number of the element unnilennium is

- (A) 109
- (B) 119
- (C) 102
- (D) 108

Q4 JEE Main 2020 - 3 September (Evening)

Among the statements (I – IV), the correct ones are

- (I) Be has smaller atomic radius compared to Mg.
- (II) Be has higher ionization enthalpy than Al.
- (III) Charge/radius ratio of Be is greater than that of Al.
- (IV) Both Be and Al form mainly covalent compounds.

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Chemistry

(A) (I), (III) and (IV)

(B) (I), (II) and (IV)

(C) (1), (II) and (III)

(D) (II), (III) and (IV)

Q5 JEE Main 2020 - 3 September (Evening)

The five successive ionization enthalpies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol⁻¹.

The number of valence electrons in the element is

(A) 3

(B) 4

(C) 2

(D) 5

Q6 JEE Main 2020 - 4 September (Morning)

The elements with atomic numbers 101 and 104 belong to, respectively,

(A) Group 6 and Actinoids

(B) Actinoids and Group 4

(C) Group 11 and Group 4

(D) Actinoids and Group 6

Q7 JEE Main 2020 - 4 September (Morning)

The ionic radii of O²⁻, F⁻, Na⁺ and Mg²⁺ are in the order

(A) F⁻ > O²⁻ > Na⁺ > Mg²⁺

(B) Mg²⁺ > Na⁺ > F⁻ > O²⁻

(C) O²⁻ > F⁻ > Mg²⁺ > Na⁺

(D) O²⁻ > F⁻ > Na⁺ > Mg²⁺

Q8 JEE Main 2020 - 4 September (Evening)

The process that is NOT endothermic in nature is

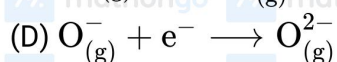
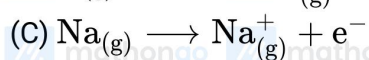
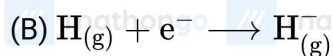
(A) $\text{Ar}_{(g)} + e^- \longrightarrow \text{Ar}_{(g)}^-$

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Chemistry



Q9 JEE Main 2020 - 5 September (Morning)

The difference between radii of 3rd and 4th orbits of Li^{2+} is ΔR_1 . The difference between the radii of 3rd and 4th orbits of He^{+} is ΔR_2 . Ratio $\Delta R_1 : \Delta R_2$ is

(A) 03:02

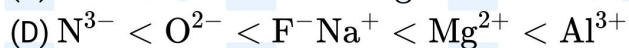
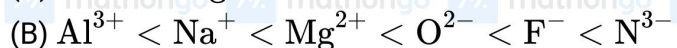
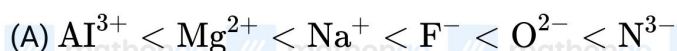
(B) 08:03

(C) 02:03

(D) 03:08

Q10 JEE Main 2020 - 5 September (Evening)

The correct order of the ionic radii of O^{2-} , N^{3-} , F^{-} , Mg^{2+} , Na^{+} and Al^{3+} is



Q11 JEE Main 2020 - 6 September (Morning)

The set that contains atomic numbers of only transition elements, is

(A) 21,32,53,64

(B) 9,17,34,38

(C) 37,42,50,64

(D) 21,25,42,72

Q12 JEE Main 2020 - 6 September (Evening)

The atomic number of Unnilunium is

Q13 JEE Main 2020 - 7 January (Morning)

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Periodic Properties

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Chemistry

The electron gain enthalpy (in kJ/mol) of fluorine, chlorine, bromine and iodine, respectively are :

- (A) $-296, -325, -333, -349$
- (B) $-333, -349, -325, -296$
- (C) $-349, -333, -325, -296$
- (D) $-325, -333, -349, -296$

Q14 JEE Main 2020 - 7 January (Morning)

Atomic radius of Ag is similar to

- (A) Cu
- (B) Hg
- (C) Au
- (D) Ni

Q15 JEE Main 2020 - 7 January (Evening)

Which one of the following amongs each pair will release maximum energy on gaining one electron

$(A = F, Cl), (B = S, Se), (C = Li, Na)$

- (A) $(A) = Cl, (B) = S, (C) = Li, (Na)$
- (B) $(A) = F, (B) = Se, (C) = Li$
- (C) $(A) = F, (B) = Se, (C) = Na$
- (D) $(A) = Cl, (B) = S, (C) = Na$

Q16 JEE Main 2020 - 8 January (Morning)

The third ionization enthalpy is minimum for :

- (A) Mn
- (B) Co
- (C) Fe
- (D) Ni

Q17 JEE Main 2020 - 8 January (Morning)

The first ionization energy (in kJ/mol) of Na, Mg, Al, Si in $kJ mol^{-1}$ respectively are:

- (A) $786, 737, 577, 496$

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Chemistry

(B) 497, 577, 737, 786

(C) 786, 739, 577, 497

(D) 739, 577, 786, 487

Q18 JEE Main 2020 - 8 January (Evening)

The increasing order of the atomic radii of the following elements is :

(a) C

(b) O

(c) F

(d) Cl

(e) Br

(A) $Br > Cl > C > O > F$

(B) $Br < Cl < C < O < F$

(C) $Cl < C < O < F < Br$

(D) $C > F > O > Br > Cl$

Q19 JEE Main 2020 - 9 January (Morning)

B has a smaller first ionization enthalpy than Be . Consider the following statements :

(I) It is easier to remove $2p$ electron than $2s$ electron

(II) $2p$ electron of B is more shielded from the nucleus by the inner core of electrons than the $2s$ electrons of Be

(III) $2s$ electron has more penetration power than $2p$ electron

(IV) atomic radius of B is more than Be

(atomic number $B = 5$, $Be = 4$)

The correct statement are :

(A) (i), (ii), (iii), (iv)

(B) (i), (iii), (iv)

(C) (ii), (iii), (iv)

(D) (i), (ii), (iii)

Answer Key

Q1 (D)

Q2 (A)

Q3 (A)

Q4 (B)

Q5 (A)

Q6 (B)

Q7 (D)

Q8 (B)

Q9 (C)

Q10 (A)

Q11 (D)

Q12 (101)

Q13 (B)

Q14 (C)

Q15 (A)

Q16 (C)

Q17 (A)

Q18 (A)

Q19 (D)

Q1

Atomic radius decreases on moving left to right in periodic table, while other three properties given increases (in magnitude) on moving left to right across a period.

Q2

In periodic table in a period on moving from left to right as atomic number increases the nature of oxide follows the order basic, amphoteric and acidic i.e., oxide of X is basic, oxide of Y amphoteric oxide of Z is acidic Hence atomic number follows the order $z > Y > x$

Q3

For Unnilennium

IUPAC symbol - Uue

Atomic No. -109

Q4

$\text{Be} < \text{Mg}$ (atomic radius)

$\text{Be} > \text{Al}$ (1. E_1)

$\text{Be} \simeq \text{Al}$ (Charge/radius ratio)

Both Be and Al form mainly covalent compound.

Q5

There is a sudden jump after 3^{rd} 1. E. due to attainment of noble gas configuration. So, the number of valence electrons in this element are 3.

Q6

Actinoids contains 14 elements with atomic number 90 to 103. Hence element with atomic number 101 is Actinoids.

Element with atomic number 104 is a d-block element of group 4

Q7

Periodic Properties

Hints & Solutions

JEE Main 2020 Chapterwise

Chemistry

Among isoelectronic species, greater the Z_{eff} smaller will be the radius.

Order of Z_{eff} : $\text{O}^{2-} < \text{F}^- < \text{Na}^+ < \text{Mg}^{2+}$

Order of Ionic Radii : $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+}$

Q8

$\text{Ar(g)} + e^- \longrightarrow \text{Ar}^-(\text{g})$ (Endothermic)

$\text{H(g)} + e^- \longrightarrow \text{H}^-(\text{g})$ (exothermic)

$\text{Na(g)} \longrightarrow \text{Na}^+(\text{g}) + e^-$ (endothermic)

$\text{O}^-(\text{g}) + e^- \longrightarrow \text{O}^{2-}(\text{g})$ (endothermic)

- EGE of H(g) is negative while that of Ar(g) is positive due to ns^2np^6 configuration.

2nd EGE is always positive for an atom

- Ionization potential of an atom is positive

Q9

$$r_n = a_0 \frac{n^2}{Z}$$

$$r_n \propto \frac{n^2}{Z}$$

$$\therefore \frac{\Delta R_1}{\Delta R_2} = \frac{Z_{\text{He}^+}}{Z_{\text{Li}^{2+}}} = \frac{2}{3}$$

Q10

For isoelectronic species, as the no. of protons increases, size of ions decreases.

$\therefore$ Correct order of ionic radii $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+ < \text{F}^- < \text{O}^{2-} < \text{N}^{3-}$

Q11

Elements with atomic number 21, 25, 42 and 72 belongs to transition metals.

Q12

Unnilunium

IUPAC symbol = Unu

Atomic no. (Z) = 101

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Q13

Self-Explanatory

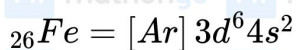
Q14

Self-Explanatory

Q15

Theory Based

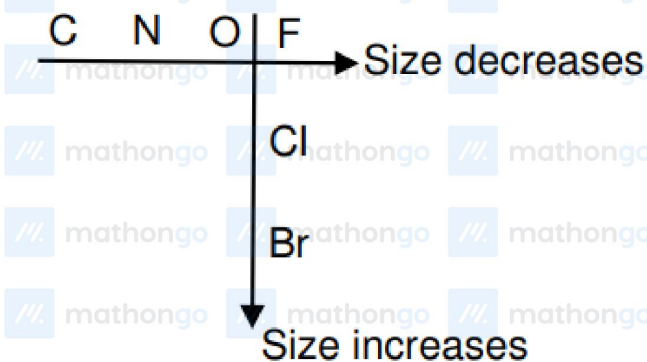
Q16



Q17

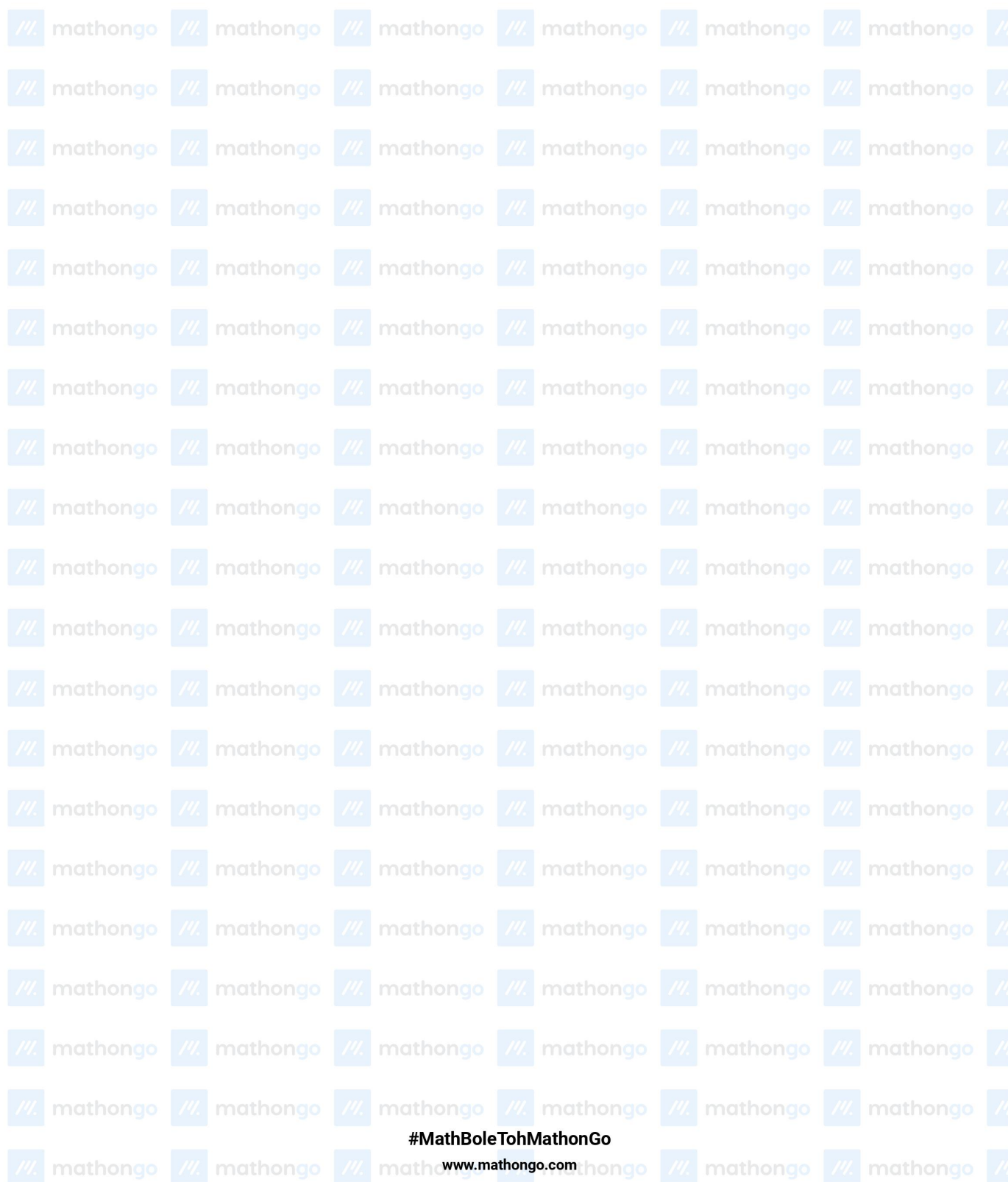
Correct order of ionization energy will be : $\text{Na} < \text{Al} < \text{Mg} < \text{Si}$

Q18



Q19

Theory Based.



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